Certified lower bounds for $\lambda_1 \cdot \text{Area}$ on cells of pentagon shape space by piecewise-affine transplantation

Computer-assisted proof candidate (for the stated cells), not peer reviewed

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Abstract

For a planar domain Ω let $F(\Omega) = \lambda_1(\Omega)\,|\Omega|$, where λ_1 is the first Dirichlet eigenvalue of the Laplacian. The Pólya–Szegő conjecture asserts that among pentagons the regular one minimises F; its value is $F_* \approx 18.9191043$. We describe a method that certifies $F(P) > F_*$ simultaneously for *every* pentagon P in a box ("cell") of a six-dimensional shape chart, using only the first eigenvalues of the 64 corner pentagons of the box and explicit matrix inequalities obtained from a piecewise-affine transplantation onto a fixed reference fan. Everything is evaluated in ball arithmetic. We present six certificates for five distinct parameter boxes at three centre locations (two certificates apply different methods to the same box), including the cell identified as hardest in a publicly available outward-continuation prototype [5], and a partial independent second-party replay of certificate components (Section 7): for boxes 1 and 3 (certificates A and B) every load-bearing component – geometry factors, all 128 corner eigenvalue bounds and the regular-pentagon bound – has been recomputed by independent code. This is a method and a set of cell certificates; it is *not* a proof of the conjecture, nor of any statement about a neighbourhood of the regular pentagon.

1 Background

Pólya and Szegő proved that the equilateral triangle and the square minimise F among triangles and quadrilaterals, $F \geq 4\pi^2/\sqrt{3}$ and $F \geq 2\pi^2$ respectively, and conjectured the same for n-gons with $n \geq 5$ [1]. Bogosel and Bucur reduced the case of fixed n to a finite (if large) computation and proved local minimality of the regular pentagon and hexagon with validated computing [2, 3]; local minimality of the regular heptagon and octagon was certified in [4]; an explicit local ball around the regular pentagon and a prototype of an outward continuation were published in [5]; the conjecture is known for convex polygons with sufficiently many sides [6]. To our knowledge (two independent literature searches, October 2026) the global pentagon case remains open. The present note contributes a cell-certification tool for the region away from the regular pentagon; we make no claim of priority beyond the specific method and certificates described here.

2 Chart, reference fan and transplantation

A pentagon is $P(x)$ with vertices $V_0 = (0,0)$, $V_1 = (1,0)$, $V_2 = (x_1, x_2)$, $V_3 = (x_3, x_4)$, $V_4 = (x_5, x_6)$, counter-clockwise. F is invariant under similarities. Let P_0 be a fixed regular pentagon (binary64

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vertices taken exactly) with vertex centroid c_0, and $T_k = \operatorname{conv}(c_0, P_{0,k}, P_{0,k+1})$, $E_k = [P_{0,k} - c_0,\ P_{0,k+1} - c_0]$. For $P = P(x)$ with vertex centroid c put

$$D_k = [V_k - c,\ V_{k+1} - c]\, E_k^{-1}, \qquad J_k = \det D_k, \qquad G_k = J_k\, D_k^{-1} D_k^{-T} \qquad (k = 0, \dots, 4).$$

If P is strictly convex, c is interior, every $J_k > 0$ and the piecewise-affine map Φ equal to $y \mapsto c + D_k(y - c_0)$ on T_k is a bi-Lipschitz homeomorphism $P_0 \to P$; then

$$\lambda_1(P) = \inf_{0 \neq v \in H_0^1(P_0)} \frac{\sum_k \int_{T_k} \nabla v^T G_k \nabla v}{\sum_k J_k \int_{T_k} v^2}, \qquad |P| = \sum_k |T_k|\, J_k. \tag{1}$$

Convexity gate. The 15 determinants $\det(V_{i+1} - V_i,\, V_j - V_i)$, $j \notin \{i, i+1\}$, are affine in each coordinate of x; positivity at the 64 corners of a box therefore implies strict convexity on the whole box. (Positivity of the J_k alone does not suffice: the regular pentagram is a counterexample.)

3 The cell inequality

A cell is a box $C = \prod_i [\ell_i, h_i]$ with binary64 endpoints. Write $x(u) = \ell + (h - \ell) \odot (1 + u)/2$, $u \in [-1, 1]^6$, let x_c ($c \in \{\pm 1\}^6$) be the corners and $\alpha_c(u) = \prod_i (1 + c_i u_i)/2$ the product (multilinear) weights; $D_{c,k}, J_{c,k}, G_{c,k}$ denote the corner quantities.

Lemma 1 (exact interpolation)**.** *$D_k(x(u)) = \sum_c \alpha_c(u) D_{c,k}$ and $J_k(x(u)) = \sum_c \alpha_c(u) J_{c,k}$.*

Proof. D_k is affine in x and multilinear interpolation reproduces affine functions. J_k is a quadratic polynomial in x whose pure squares vanish: the derivative of D_k in a single coordinate x_i moves one coordinate of one vertex and has rank ≤ 1, so its determinant is 0; hence J_k is multiaffine and equals its multilinear interpolant. □

Lemma 2 (covariance identity)**.** *With $X_c = D_{c,k}^{-1}$, $w_c = \alpha_c J_{c,k}/J_k$, $M = \sum_c w_c X_c$ and $S_k = \sum_c \alpha_c G_{c,k}$,*

$$G_k = S_k - J_k\, \mathcal{V}_k, \qquad \mathcal{V}_k = \sum_c w_c (X_c - M)(X_c - M)^T \succeq 0.$$

Proof. For 2×2 matrices the adjugate is linear, so $\operatorname{adj} D_k = \sum_c \alpha_c \operatorname{adj} D_{c,k} = J_k M$; then $G_k = \operatorname{adj} D_k (\operatorname{adj} D_k)^T / J_k = J_k M M^T$ and $\sum_c \alpha_c G_{c,k} = J_k \sum_c w_c X_c X_c^T$. □

Lemma 3 (matrix Efron–Stein)**.** *Let $G_l \succ 0$ satisfy $G_l \preceq G_{c,k}$ for all corners, $J_k^{\max} = \max_c J_{c,k}$, $p_i q_i = (1 - u_i^2)/4$, and for each coordinate i let $\Delta_i^{(e)}$ be the 32 edge differences $X_c - X_{c'}$ (c, c' differing in coordinate i), R_i their mean and $d_i^{(e)} = \Delta_i^{(e)} - R_i$. Then for every $\tau > 0$*

$$J_k \mathcal{V}_k \preceq \epsilon_k(u)\, G_l \preceq \epsilon_k(u)\, S_k,$$

$$\epsilon_k(u) = J_k^{\max} \Big[(1 + \tau)\, \lambda_{\max}\Big(G_l^{-1/2} \sum_i p_i q_i R_i R_i^T G_l^{-1/2} \Big) + \big(1 + \tfrac{1}{\tau}\big) \sum_i p_i q_i \max_e \lambda_{\max}\big(d_i^{(e)T} G_l^{-1} d_i^{(e)} \big) \Big].$$

Proof. $J_k \mathcal{V}_k \preceq \sum_c \alpha_c J_{c,k} (X_c - \bar{X})(X_c - \bar{X})^T \preceq J_k^{\max} \operatorname{Var}_\alpha(X)$ for the α-mean $\bar{X}$. The Efron–Stein inequality for the product measure α, applied to every scalar projection, gives the Loewner bound $\operatorname{Var}_\alpha(X) \preceq \sum_i p_i q_i\, \mathbb{E}_\alpha[\Delta_i \Delta_i^T]$, and $(R + d)(R + d)^T \preceq (1 + \tau) R R^T + (1 + \tau^{-1}) d d^T$ [13]. □

Choice of G_l. For each sector let $G_{\mathrm{cen}} = \frac{1}{64}\sum_c G_{c,k}$. Two admissible candidates are computed in ball arithmetic: (a) $G_{\mathrm{cen}} - \eta I$ with $\eta \geq \max_c \|G_{c,k} - G_{\mathrm{cen}}\|_2$, if it is positive definite; (b) $\mu_l G_{\mathrm{cen}}$ with $0 < \mu_l \leq \min_c \lambda_{\min}(G_{\mathrm{cen}}^{-1/2} G_{c,k} G_{\mathrm{cen}}^{-1/2})$ (the smallest root of $\det(G_{c,k} - \mu G_{\mathrm{cen}}) = 0$, rounded down). Both satisfy $G_l \preceq G_{c,k}$ for every corner; the certificate uses the one with the larger $\lambda_{\min}(G_{\mathrm{cen}}^{-1} G_l)$ (a choice, not a bound), and asserts $G_l \succ 0$. (μ_l is unrelated to the quantity μ of Section 4.) The value of τ is the best of a fixed finite list.

Theorem 4 (cell bound)**.** *Let $F_c^L \leq F(P(x_c))$ be certified corner lower bounds, $\lambda_c^L = F_c^L/|P(x_c)|$, and suppose the convexity gate holds on C and $\epsilon_k(u) < 1$. Then for every $u \in [-1,1]^6$*

$$F(P(x(u))) \;\geq\; \Big(1 - \max_k \epsilon_k(u)\Big) \min_k |P(x(u))| \, \frac{\sum_c \alpha_c(u)\, \lambda_c^L\, J_{c,k}}{J_k(x(u))}. \tag{2}$$

Proof. By Lemmas 2–3, $G_k \succeq (1-\epsilon_k) S_k$. For $v \in H_0^1(P_0)$ the numerator of (1) is $\geq \sum_k (1-\epsilon_k) \sum_c \alpha_c N_{c,k}(v) \geq (1 - \max_k \epsilon_k) \sum_c \alpha_c N_c(v)$, where $N_c = \sum_k N_{c,k}$ is the numerator for the corner $P(x_c)$, and $N_c(v) \geq \lambda_c \sum_k J_{c,k} \int_{T_k} v^2$. The denominator is $\sum_k J_k \int_{T_k} v^2$ by Lemma 1; a ratio of positive sums is at least the smallest termwise ratio. (The corner Rayleigh inequality holds only for the sum over sectors, which is why the factor is $1 - \max_k \epsilon_k$ and not paired sector by sector.) □

Exact-covariance alternative. By Lemma 2, $S_k - G_k \succeq 0$, so

$$e_k(u) = \lambda_{\max}\big(S_k^{-1/2}(S_k - G_k)S_k^{-1/2}\big) \in [0, 1)$$

satisfies $G_k \succeq (1 - e_k) S_k$ directly; here S_k and $\operatorname{adj} D_k$ (hence $G_k = \operatorname{adj} D_k (\operatorname{adj} D_k)^T / J_k$) are multiaffine. Any upper bound of ϵ_k or e_k may be used in place of ϵ_k in Theorem 4.

Two ways of using (2) are implemented.

Min-corner form. Write $f_k = J_k/|P|$ and $I(f_k) = \sum_c \alpha_c f_k(x_c)$, so that with $\lambda_c^L \geq F_{\min}/|P(x_c)|$, $F_{\min} = \min_c F_c^L$, the k-th term of (2) is at least $F_{\min}\, I(f_k)/f_k$. Tensor-linear interpolation gives $|I(f_k) - f_k| \leq \mathrm{err}_k = \sum_i \frac{(h_i - \ell_i)^2}{8} \sup_C |\partial_i^2 f_k|$ with the full side lengths $h_i - \ell_i$, the supremum being enclosed on 64 sub-boxes whose union is C (interval evaluation, asserted coverage). With $f_k^{\min}$ a positive lower bound of f_k on C and ϵ_k evaluated at $p_i q_i = 1/4$ (its maximum over u),

$$F(P) \;\geq\; F_{\min} \cdot \min_k (1 - \epsilon_k) \cdot \Big(1 - \max_k \frac{\mathrm{err}_k}{f_k^{\min}}\Big) =: \mathrm{factor} \cdot F_{\min} \quad \text{on all of } C,$$

provided all $J_{c,k} > 0$, $f_k^{\min} > 0$, $\epsilon_k < 1$ and $\mathrm{err}_k < f_k^{\min}$ (all asserted). The code also multiplies by $J_{\min}/(J_{\min} + \delta)$ with $\delta = \sum_i (\frac{h_i - \ell_i}{2})^2 |\det \partial_i D_k|$; by Lemma 1 $\delta = 0$, so this factor is 1.

T1 form. Since $Q_k > 0$ and $\epsilon < 1$, the right-hand side of (2) equals $\min_{k,k'} Q_k(u)\,(1 - \epsilon_{k'}(u))$ with $Q_k = |P| \sum_c \alpha_c \lambda_c^L J_{c,k}/J_k$. It is bounded below over $u \in [-1,1]^6$ by branch and bound: on a sub-box with centre c and radii r, each of the 25 products is bounded below by $f(c) - \sum_i |\partial_i f(c)| r_i - \frac{1}{2}\sum_{i,j} \sup_{\mathrm{box}} |\partial_{ij} f|\, r_i r_j$ (point jet at the centre, interval Hessian over the box); multiaffine pieces have exact ranges from the sub-box corner values. For ϵ_k and e_k the $\lambda_{\max}$ of a positive semidefinite 2×2 matrix is replaced by the polynomial majorant $\mathrm{tr}/2 + (q + s^2)/(2s)$, $q = \mathrm{tr}^2/4 - \det$ (AM–GM, $s > 0$ fixed per box), or by the trace. Cells C, D and F use the Efron–Stein bound only; cell E uses, per sub-box and sector, the smaller of the valid majorants of ϵ_k and e_k. The accepted leaves form an exact dyadic partition of $[-1,1]^6$ and are part of the certificate.

4 Corner eigenvalues and the regular pentagon

Corner lower bounds use the method of particular solutions: $u = \sum_{j,m} c_{jm} J_{\nu_{jm}}(kr_j)\sin(\nu_{jm}\theta_j)$, $\nu_{jm} = m\pi/\alpha_j$, solves $\Delta u + \tilde{\lambda} u = 0$ exactly ($k^2 = \tilde{\lambda}$); by Fox–Henrici–Moler some eigenvalue satisfies $|\tilde{\lambda} - \lambda_n| \le \mu\lambda_n$ with $\mu = |P|^{1/2}\sup_{\partial P}|u|/\|u\|_2$ [7, 8]; $\|u\|_2$ is bounded below through the Helmholtz mean-value identity on an inscribed disk; and, when $\mu < 1$, $\tilde{\lambda}|P|/(1-\mu) < 2\pi j_{0,1}^2 \le \lambda_2|P|$ (Hong–Krahn–Szegő [9, 10, 11]) identifies $\lambda_n = \lambda_1$. Hence $\lambda_1 \ge \tilde{\lambda}/(1+\mu)$. All 384 corners have $\mu \le 3.1 \cdot 10^{-4}$. Every corner certificate records the vertices, $\tilde{\lambda}$ (a binary64 number), μ and the 40 coefficients, so that a verifier evaluates the *same* explicit function. The producer verifier treats the stored enclosures of μ and of the area as certificate data: it re-checks $\mu < 1$, the index inequality and $\tilde{\lambda}/(1+\mu)$ from them, but does not recompute the boundary supremum or the disk bound; those are recomputed by the second-party checker for the corners listed in Section 7. All corner pentagons are strictly convex (convexity gate), so every θ_j is taken in $[0, \alpha_j]$ with $\alpha_j < \pi$ and the branch cut of θ_j lies outside $\overline{P}$. The stored binary64 vertex coordinates, $\tilde{\lambda}$ and coefficients define the trial function exactly; interval arithmetic encloses its evaluations, and the final reported lower and upper bounds are rounded outward to binary64.

The regular pentagon is bounded from above in two independent ways: (i) the FHM upper bound $\tilde{\lambda}/(1-\mu)$ at a binary64 regular pentagon P_t whose copy shrunk by the factor $1 - 10^{-12}$ lies inside the exact regular pentagon (checked in ball arithmetic), with domain monotonicity, giving $F_* \le 18.919134195733918$; (ii) a conforming P_2 finite-element Rayleigh quotient on an inscribed rational pentagon with directed-rounding exact integration (second party; source and logs in `second_party/p2_fem`), giving $F_* \le 18.919118220170337$. The trial vector of that logged run is not archived, so this value is a historical log that cannot be replayed digit for digit from the package; rerunning the two scripts certifies a fresh upper bound whose last digits may differ. All margins in this note are computed against the larger value (i), $F_*^{\mathrm{up}} = 18.919134195733918$. Validated eigenvalue bounds on polygons by other means are available in [12].

5 Results

There are six certificates (A–F) for five distinct boxes (1–5) at three centres. Each box is $C = \prod_i [\ell_i, h_i]$ with the binary64 endpoints listed in its certificate file (centre $\pm\rho$ in every coordinate, sup norm); x_* is the regular pentagon in the chart. "Margin is the certified lower bound of F on the whole box minus $F_*^{\mathrm{up}} = 18.919134195733918$. Boxes 1 and 2 share a centre, the cell identified as hardest in the outward-continuation prototype of [5], and $1 \subset 2$; boxes 3 and 4 share a centre at chart distance 0.1 and $3 \subset 4$; certificates B and F certify the same box 3 by the two methods; box 5 is centred at the hardest of 480 random centres of our own cost sweep. The nested and repeated certificates are kept as method comparisons.

cert.	box	form	$\|x_C - x_*\|_2$	ρ	certified lower bound	margin	B&B leaves
A	1	min-corner	0.5736	0.025	18.976746833	5.76e-02	–
B	3	min-corner	0.1003	0.012	18.920850256	1.72e-03	–
C	2	T1	0.5736	0.03	18.919213342	7.91e-05	1179
D	4	T1	0.1003	0.015	18.919134372	1.77e-07	3368
E	5	T1	0.5738	0.0239	18.919156378	2.22e-05	3335
F	3	T1	0.1003	0.012	18.919185215	5.10e-05	220

For boxes 4 and 5 the min-corner form fails (its bound is 18.9099 resp. 18.6492) and only the T1 form certifies the box; box 5 needs the exact covariance majorant e_k in addition to Lemma 3.

6 Cost of a cover (estimate)

To estimate the cost of covering a whole region we measured, at 480 centres log-uniformly distributed in chart distance $r \in [0.02, 0.6]$ (directions uniform), for which half-widths ρ the rigorous T1 branch and bound succeeds (a search on a grid of ratio 0.85 up to the largest passing size). In this measurement the corner eigenvalues are floating-point values reduced by a relative margin $2 \cdot 10^{-5}$ (a strict corner certificate loses about that much), so the numbers below are an *estimate*, not a certificate.

A proof needs only one labelling of each pentagon. We use the Dirichlet domain D of the relabelling action of D_5: the points of the chart whose own labelling is (weakly) the closest of their ten labellings to x_*. The action is not an isometry of the chart (at x_* the Jacobian of a cyclic relabelling has singular values $2.66, 2.66, 1, 1, 0.375, 0.375$), so D does not occupy a tenth of a shell around x_*; 343 of the 480 centres lie in D. The shell $a \le r < b$ has measure $\pi^3 r^5\, dr$. Let $\rho(x) = q(x)\, r$ be the half-width chosen at x by the rule below. For r log-uniform on $[a, b)$ and uniform directions

$$N_D = \frac{\pi^3}{64} \ln \frac{b}{a}\, \mathbb{E}\big[\mathbf{1}_D\, q^{-6}\big] = \int_{D \cap \{a \le r < b\}} (2\rho(x))^{-6}\, dx,$$

and the mean over all sampled centres of the band, with those outside D counted as zero, is an unbiased Monte Carlo estimate of this integral. The integral is a continuum proxy for the number of cells, not the number of boxes of a constructed rigorous cover: cells that cross the boundary of D, the interfaces of an adaptive grid, overlaps and the certification overhead are not counted. The cost is estimated in the same way, with the integrand multiplied by the cost per cell: 1.5 amortised strict corners (19.5 CPU seconds each, shared between neighbouring grid cells) plus the branch-and-bound time measured at that size. For each centre we use the tested passing size with the smallest cost per volume; smaller sizes were not measured and may be cheaper. Times are CPU seconds on the author's laptop (Intel i9-14900HX, WSL2); the intervals are 95% bootstrap intervals over the centres of each band.

band of r	centres	in D	median ρ/r in D	cells (proxy)	s/cell	CPU-days (95%)
$[0.02, 0.05)$	142	106	0.150	$3.0 \cdot 10^7$	93	32521 (261–92778)
$[0.05, 0.20)$	209	146	0.151	$9.8 \cdot 10^4$	323	367 (277–480)
$[0.20, 0.60)$	129	91	0.115	$2.9 \cdot 10^6$	114	3807 (998–8975)

For $r \in [0.05, 0.6)$ this gives a proxy of about $3.0 \cdot 10^6$ cells and 4174 CPU-days (bootstrap interval 1350–9305). The estimate rests on few shapes: one centre ($r = 0.574$, one edge about 3.3 times shorter than the longest) carries 63% of the band $[0.2, 0.6)$, and without it the total would be about 1792 CPU-days. Shrinking every cell to 0.8 times its largest passing size, with a branch-and-bound time proportional to ρ^β ($\beta = 11.3$–14.2, the interquartile range fitted on pairs of passing sizes by the second party), gives 3885–5025 CPU-days, so a uniform change of the size does not help. Including $[0.02, 0.05)$ the estimate is $3.3 \cdot 10^7$ cells and 36694 CPU-days, but that band is dominated by one centre ($r = 0.029$, 86% of the band) and its interval (261–92778) is too wide to be useful. The region $r \ge 0.6$ and a neighbourhood $r < 0.02$ of the regular pentagon are not included, and a different fundamental domain, or a choice of labelling per region, could be cheaper; this was not measured.

3 of the 480 centres did not pass at any tested size. All of them lie at $r < 0.025$ and outside D (their closest labellings are at $r = 0.013$–0.018), and every failed test failed because the corner values, reduced by the $2 \cdot 10^{-5}$ margin, were already below F_*^{up}. This concerns the tested sizes and the chosen margin, not every possible refinement. Lying outside D, these centres contribute zero

to the estimate; a centre in D that passed at no tested size would leave the estimate undefined, and the estimator stops in that case. Close to the regular pentagon the accuracy of the corner eigenvalues, not the transplantation bound, limits the method.

7 Independent second-party replay

A second party (GPT, OpenAI) wrote independent checkers that share no source with the producer and never executed the producer code. "PASS" means that the independent bound is at least as strong as the one used by the producer certificate. The table separates checks of saved certificate data from independent recomputation.

item	status	method of the second party
Upper bound F_*^{up} for the regular pentagon	PASS	conforming P_2 FEM Rayleigh quotient on an inscribed rational pentagon, exact integration with directed rounding: $F_* \le 18.919118220170337$ (source and logs: `second_party/p2_fem`; the trial vector of the logged run is not archived, so the value is a historical log and a rerun certifies a fresh bound)
Convexity gate (all six certificates)	PASS	exact rational corner determinants
B&B leaves form an exact dyadic partition of $[-1,1]^6$ (C–F); coverage only	PASS (8,102 leaves)	exact dyadic tree reconstruction
Internal consistency of the 384 saved corner records ($\mu < 1$, index inequality)	PASS	rational re-parsing of all saved balls; a check of saved data, not a recomputation
Min-corner geometry factor, certificates A and B (boxes 1, 3)	PASS 2/2	exact `Fraction` recomputation (factor $0.995193 \ge 0.995121$, $0.999346 \ge 0.999344$)
Corner eigenvalue lower bounds, certificates A and B (boxes 1, 3) (128 corners)	PASS 128/128	own interval Bessel series with tail bounds, third-order Taylor boundary bound, disk L^2 bound, FHM, Krahn–Szegő index, on the exported coefficients; results in `second_party/results/`; smallest margin over the producer bound 2.4e-05
Corner eigenvalue lower bounds, worst corner of each of C, D, E	PASS 3/3	as above
Whole-leaf lower bounds of the T1 form (C–F)	not yet replayed	not done: an independent second-order Taylor implementation reproduces the centre values and gradients of Q_k (220 leaves of F, to $4 \cdot 10^{-15}$) but does not yet bound whole leaves

Boxes 1 and 3 (certificates A and B) are fully replayed by the second party: geometry factors (exact fractions), all 128 corner eigenvalue lower bounds and the regular-pentagon upper bound (from a logged run whose trial vector is not archived). Certificates C–F are producer certificates

whose whole-leaf bounds have not yet been replayed independently, and should be regarded accordingly.

8 Limitations

The certified boxes are small (side $2\rho \le 0.06$) at chart distance 0.1–0.57 from the regular pentagon; no statement is made about the region between them, about a neighbourhood of the regular pentagon, about non-convex pentagons, or about the conjecture. Near the regular pentagon the gap $F - F_*$ shrinks quadratically and the corner-eigenvalue accuracy, not the transplantation bound, limits the cell size. The arithmetic relies on Arb/python-flint [14] (producer) and on mpmath interval arithmetic and Python's `Fraction`/`Decimal` (second party); no formal proof assistant was used.

9 Roles of AI systems

Claude (Anthropic) designed the method, wrote and revised the producer algorithms and code, and interpreted the resulting logs. GPT (OpenAI) did the literature review and the mathematical audit and wrote the independent second-party checkers (Section 7), whose code shares no source with the producer. The code ran as local Python processes, in the AI systems' sandboxes and on the author's computer; the author initiated and coordinated the runs and decided on the release. The author did not establish the mathematical correctness by hand; responsibility for the publication and its correctness lies with the author. Errors, if any, are reported in the repository and corrected in later versions.

Data and code

All certificates, producer code, independent checkers, replay commands and a SHA-256 manifest are in the accompanying repository (CC BY 4.0 for text and data, MIT for code).

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